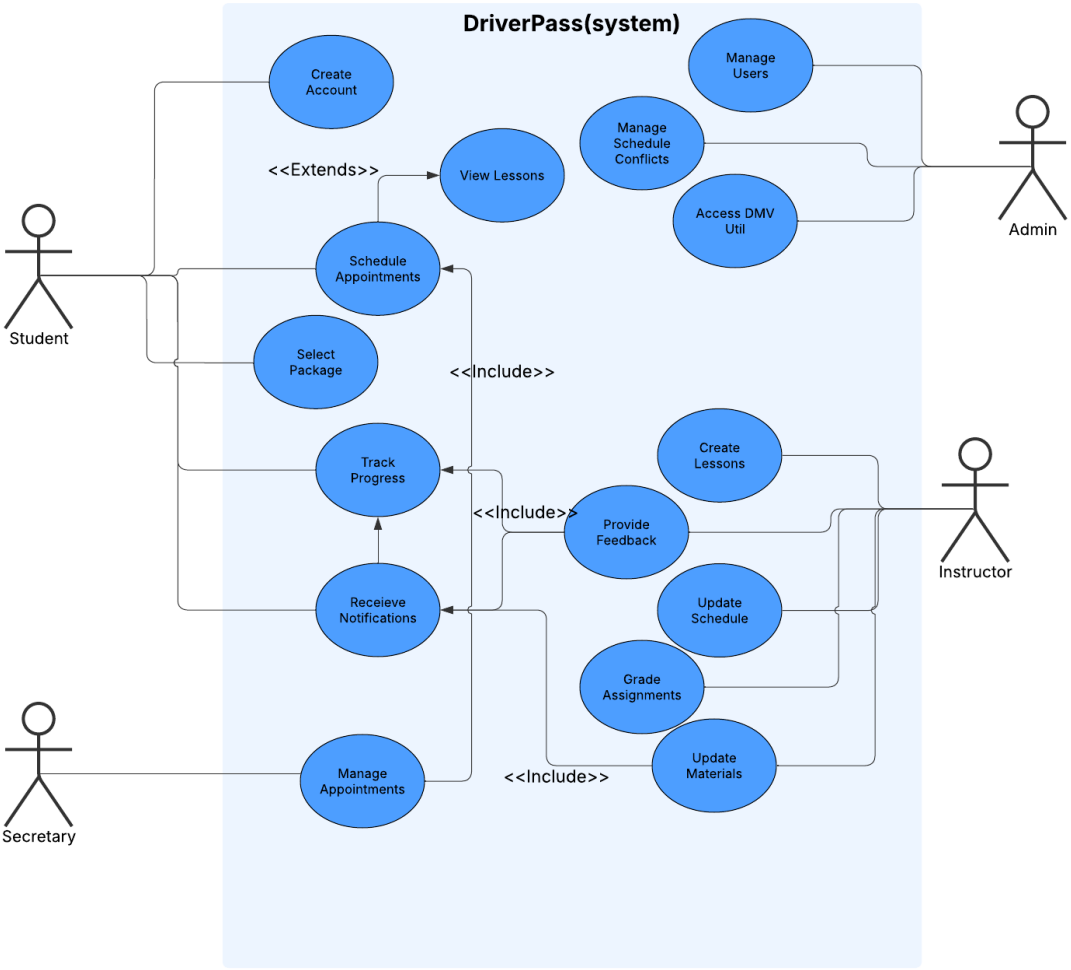
CS 255 System Design Document Template

This template lays out all the different sections that you need to complete for Project Two. Each section has guidance to prompt your thinking. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead the goal is to complete each section based on what your client's needs are. Remove this note when you are finished, and replace all bracketed text with the relevant information.

UML Diagrams

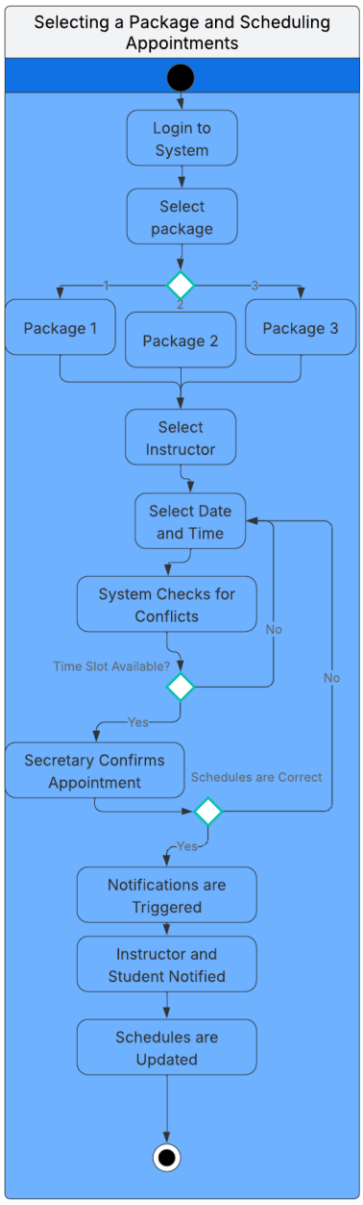
UML Use Case Diagram

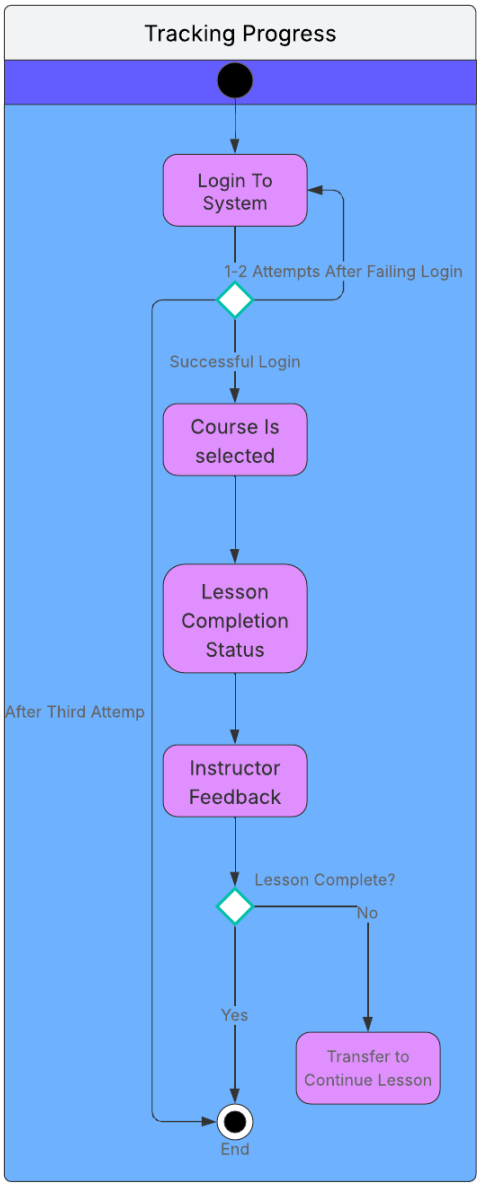


UML Activity Diagrams

*[You were asked to choose **two** use cases and create **two** activity diagrams, one for each use case. Please insert **both** of your activity diagrams here. Check to make sure that you included appropriate*

components and symbols and that your design meets the client's needs.]

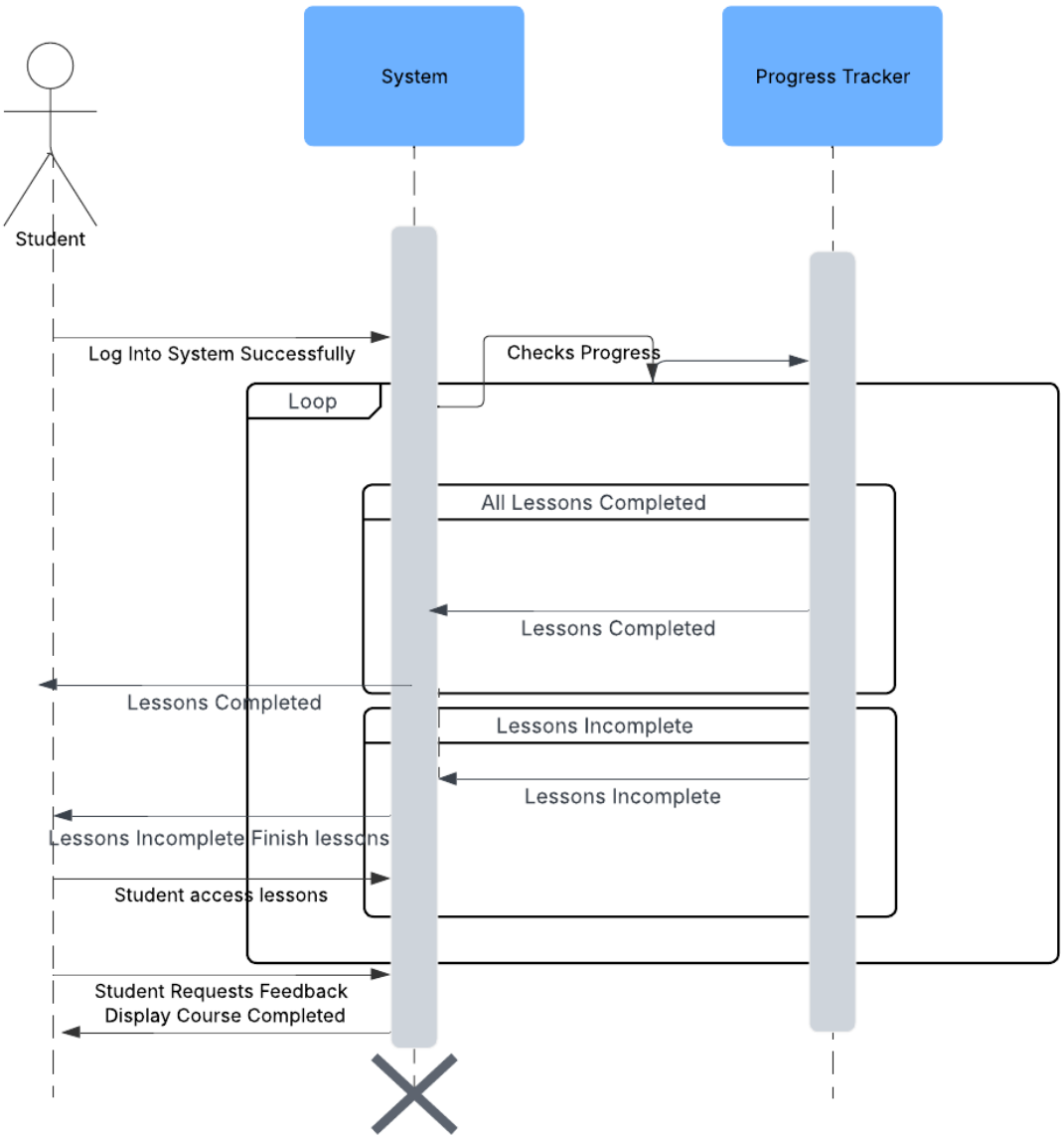




UML Sequence Diagram

*[You were asked to create a sequence diagram based on **one** of the use cases you chose. Please insert your sequence diagram here. Check to make sure that you included appropriate components and*

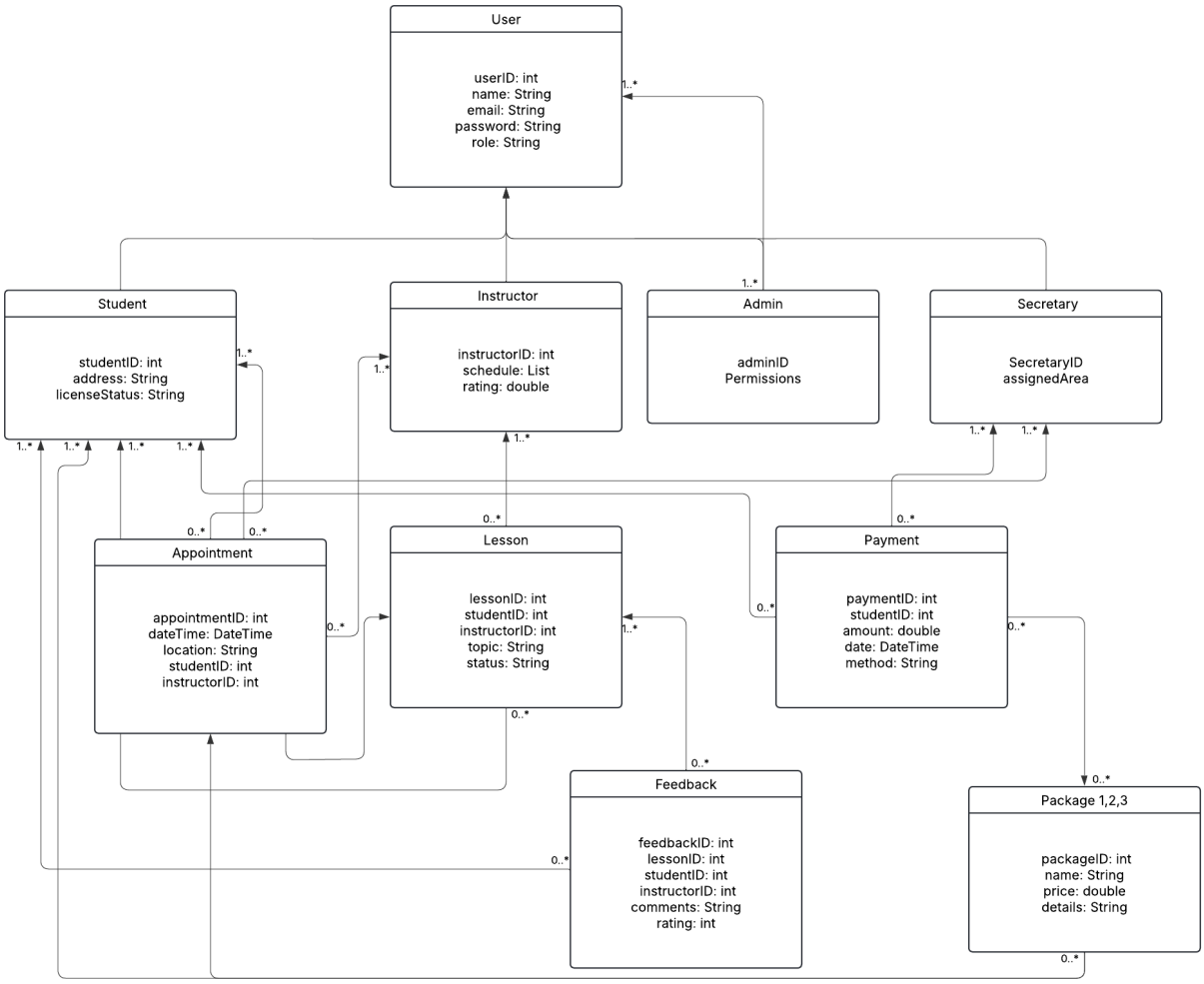
symbols and that your design meets the client's needs.]



UML Class Diagram

*[You were asked to create a class diagram based on the different classes and attributes needed for your system design. You are **not** required to include methods, but you may if you wish. Please insert your class diagram here. Check to make sure that you included appropriate components and symbols and that your*

design meets the client's requirements.]



Technical Requirements

1. Hardware Requirements
    a. Servers
        i. Cloud based servers either through AWS or Azure that help with scalability
        ii. Ram could possibly matter for these systems but foot traffic should be monitored first
        iii. 8-16gb of ram and 4-6 core cpus would be great for mid tier specs on servers

iv. SSD and HDD backed database storage, ssd for faster performance and HDD for more storage or backups

v. End user should have a web browser etc. Chrome Firefox Opera Safari and access lessons on either their own PC or phone

vi. Secretary workstations that use Windows and have MS office for scheduling and managing schedules

2. Software
    a. Backend
        i. Java Focused with Spring Boot or Node.js
        ii. MySQL for the database
        iii. RESTful based API
    b. Frontend
        i. React.js focused for UI
        ii. Mobile and PC friendly therefore needs to be developed a bit more closely with backend to ensure proper management
    c. Security
        i. OAuth 2.0 for user authentication
        ii. HTTPS for data encryption
        iii. RBAC control as well for Admin Student and Secretary
3. Tools
    a. Lucidchart for UML diagrams
    b. Github for version control
    c. Docker for containerization
    d. CI/CD pipeline to focus on GItHub

e. Payment integration with PayPal API to ensure smoother transactions

f. Email service for notifications

g. DMV API for real time updates for lessons as well

4. Nonfunctional Requirements

    a. Less than five seconds fir appointment booking

    b. Support 350-500 users

    c. Have consistent uptimes with 99.9% of the time and avoid redundancy

    d. Possibly have an auto-scaling for peak times or time of year

5. Key Technical Dependencies

    a. Database schema that matches my class diagram of User table and the role column and an appointment conflict detector that ensures that a secretary and the system notify the instructor of availability and the student

    b. Real time notifications about feedback and lesson status updates

Resources

Amazon Web Services. (n.d.). *MySQL vs. PostgreSQL - Comparing Relational Database Management Systems (RDBMS) - AWS*. Amazon Web Services, Inc. https://aws.amazon.com/compare/the-difference-between-mysql-vs-postgresql/

Kumar, S. (2023, October 19). *Implementing Role-Based Access Control (RBAC) in Node.js*. Medium. https://medium.com/@techsuneel99/implementing-role-based-access-control-rbac-in-node-js-871591b80a83

Stripe. (n.d.). *Stripe API Reference*. Docs.stripe.com. https://docs.stripe.com/api

The Total Economic Impact™ Of Red Hat OpenShift Cloud Services. (2024, March 4). Redhat.com. https://www.redhat.com/en/engage/economic-impact-of-cloud-service-analyst-material?sc_cid=RHCTN0230000254314&gclsrc=aw.ds&gad_source=1&gclid=Cj0KCQjwh_i_BhCzARIsANimeoEcECFrDVdq44ajcFuUIz-fm9HYUTS21YPIPsXFQdIdruJfhV3IcnUaAq8nEALw_wcB